New York University
Department of Psychology
Center for Neural Science

Duration judgments with conflicting audiovisual cues

Ömer F. Yildiran, Long Ni, Michael S. Landy
Department of Psychology · Center for Neural Science · New York University

QR

TL;DR **Auditory duration judgments are pulled linearly toward visual duration; bias scales with auditory noise but shows no non-linear breakdown.**

11 participants · 4 tasks · 5 models · 2,156 trials/subj

01 Question

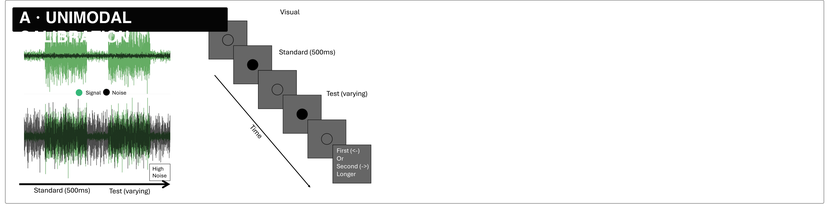
Does causal inference break audiovisual duration integration when AV conflicts grow large, the way it does in spatial localization?

- Fuse, segregate, or switch between duration cues?
- How does auditory reliability shape visual bias?
- Can behavior discriminate the computational strategies?

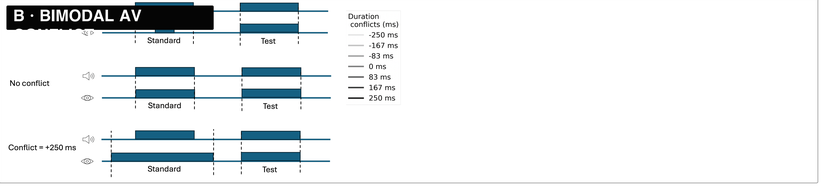
We test 7 AV-duration conflicts (±250 ms) at two auditory-noise levels and compare forced fusion, causal-inference variants, and cue switching.

02 Stimuli & Tasks

2-IFC · 500 ms standard



Auditory and visual duration-discrimination tasks calibrate per-modality sensory noise.

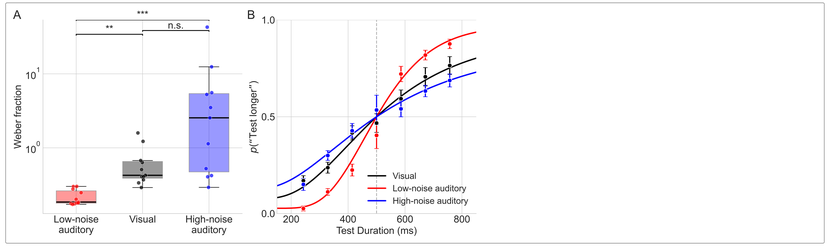


Conflict is applied only to the visual standard; observers judge which interval was longer in audition.

7	±250	2	2,156
conflicts	ms range	noise levels	trials/subj

03 Cue Reliability

Auditory reliability is highest at low noise and lowest at high noise; vision sits in between.

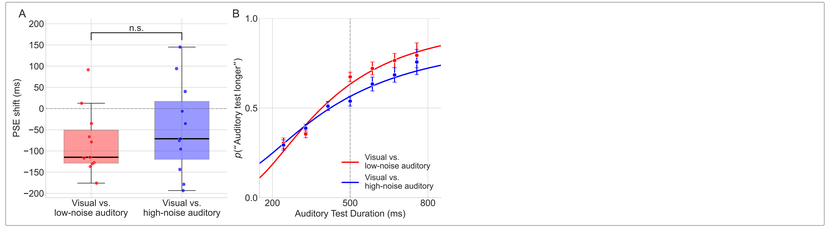


0.18	0.42	2.54
WF low-noise	WF visual	WF high-noise

04 Modality Bias

cross-modal calibration

We estimate each subject's auditory-visual bias before adding conflict, then correct the visual stimulus so nominal zero conflict is perceptually matched.

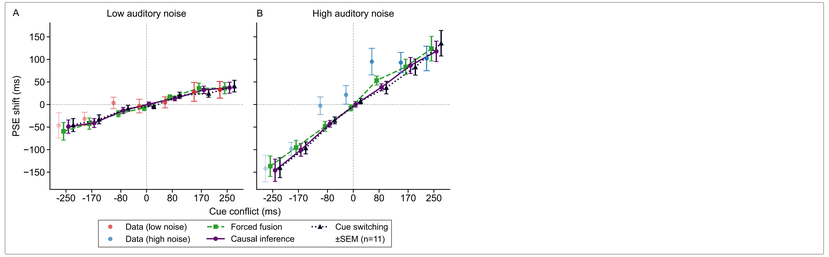


LOW-NOISE PSE SHIFT
-116 ms
W = 6, p = .014

HIGH-NOISE PSE SHIFT
-73 ms
n.s.

05 Main Result · PSE Shifts

Visual bias grows approximately linearly with conflict and is much larger under high auditory noise.

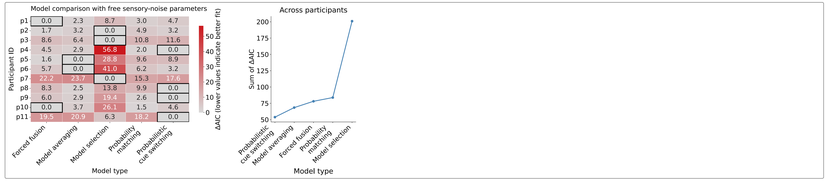


LOW-NOISE SLOPE
0.144
r = 0.91, p = .004

HIGH-NOISE SLOPE
0.606
r = 0.96, p < .001

06 Models & Comparison

free sensory noise



M1 Forced fusion
Always integrate auditory and visual cues using reliability-weighted averaging.

M2- Causal inference
Fuse or segregate according to the posterior probability of a common cause.

M5 Probabilistic cue switching
Use auditory or visual cue alone on each trial with a fixed switching probability.

Take-home: linear bias, flat model evidence, and an identifiability limit.